

Dahua ITS Integration Specification

ITSAPI

V1.22

Dahua-ITS-Integration-Protocol-(ITSAPI)_V1.22 for Asteroidea

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1 Scope

This specification is applied to the docking of Dahua intelligent transportation equipment with a third-party platform, and completes the basic equipment basic information reporting, equipment capture information and active push of picture data. This document is used by Dahua internal developers, third-party platform developers and operation and maintenance personnel.

Affirmation: Do not forward, modify or copy without permission.

2 Terms, definitions and normative references

2.1 Terms and definitions

- HTTP: HyperText Transfer Protocol
- JSON: JavaScript Object Notation
- For all business functions, the fields are concerned about the case format and the message reply content. The details are subject to the protocol definition
- Equipment basic information function, with priority higher than other business functions; When this function is enabled but the docking fails, other functions will not be pushed

2.2 Normative citation

The following normative documents are applicable to this specification.

- IETF RFC 2616 Hypertext Transfer Protocol HTTP/1. 1 (Hypertext Transfer Protocol-- HTTP/1. 1)
- IETF RFC 2617 HTTP Authentication: Basic and Digest Access Authentication.

3 Protocol framework

All communication in this specification is based on HTTP protocol, and the implementation mechanism should comply with RFC 2616. Digest authentication is supported, see chapter 4 of the specification for details.

Data transmission should support HTTP long connection and short connection;

The Http header field "Connection: Keep-Alive" is used to denote a long connection, and the Http header field "Connection: close" is used to denote the last transaction in a long connection(Transaction).

Business load adopts JSON data format, see each business definition for details; The HTTP header field "ContentType: application/json; charset=UTF-8" is used, and the encoding is uniformly UTF-8.

Note: The server side that implements this model must support IO reuse so that the supporting business can be completed on different connections.

4 Authentication

All features covered in this specification support HTTP Digest authentication, as detailed in RFC 2617. Note: Whether to enable the authentication function is determined by specific items.

Example:

The device actively sends an HTTP push request, and the platform replies 401 Unauthorized;

```
HTTP/1.1 401 Unauthorized
WWW-Authenticate: Digest realm="DH_00408CA5EA04",
nonce="000562fdY631973ef04f77a3ede7c1832ff48720ef95ad",
stale=FALSE, qop="auth"
```

The device assembles the Authorization header field according to the WWW-Authenticate information in the 401 message.

```
Authorization: Digest username="admin", realm="DH_00408CA5EA04",
nc=00000001, cnonce="0a4f113b", qop="auth",
nonce="000562fdY631973ef04f77a3ede7c1832ff48720ef95ad",
uri="/cgi-bin/magicBox.cgi?action=getLanguageCaps",
response="65002de02df697e946b750590b44f8bf"
```

5 Features

5.1 Basic information push

Basic information push is used for the platform to obtain and manage relevant equipment information. **Priority over other business functions.**

URI	/NotificationInfo/DeviceInfo			
Description	Push the basic information of equipment. After the first push is successful, push it again when the basic information changed			
Method	URI Request Parameters	Request Message	Response Message	Remarks

POST	None	Refer to Device Basic Information Message Definition	Refer to Basic information reply response definition	
Notes				

Device Basic Information Message Definition:

Sample	Description	Mandatory	Value
{	Basic Information		
"DeviceName" : "DH-ITC205"	Device Name	Yes	
"DeviceModel" : "DH-ITC205"	Device Model		
"DeviceType" : "Tollgate"	Device Type		"Tollgate": ANPR "E-Police": Electric alarm equipment
"Manufacturer" : "Dahua"	Brand Name		
"IPAddress" : "192.168.1.108"	IPv4 Address	No	
"IPv6Address" : "::"	IPv6 Address	No	
"MACAddress" : "A:B:C"	MAC Address	No	
"DeviceID" : "1c11c9c4-8dbc-4391-0dd7-56764ba18dbc"	Unique identification, using UUID	Yes	Default using UUID
}			

Basic information reply definition:

Sample	Description	Mandatory	Value
{	Basic information reply		
"Result" : true		Yes	true: Success; false: Failed;
"Message" : "Success"		Yes	Message processing result description
}			

5.2 Heartbeat Message push

Heartbeat is used by the platform to identify the status of the device. If it is not received within 3 (recommended value) cycles (recommended value: 5mins), the platform can generate an alarm to check the running status of the device.

URI	/NotificationInfo/KeepAlive
-----	-----------------------------

Description	Heartbeat message, which is used by the platform to regularly identify the equipment status			
Method	URI Request Parameters	Request message	Response message	Remarks
POST	None	Refer to Heartbeat Push Message body definition	Refer to Heartbeat response Definition	
Notes				

Heartbeat push message definition:

Sample	Description	Mandatory	Value
{	Heartbeat push		
"Active" : "keepAlive"	Heartbeat	Yes	"keepAlive": Indicates a heartbeat message
"DeviceID": "1c11c9c4-8dbc-4391-0dd7-56764ba18dbc"	Consistent with the information reported by the equipment foundation		
}			

Heartbeat reply message definition:

Sample	Description	Mandatory	Value
{	Heartbeat push		
"Active" : true	Result of heartbeat	Yes	true: Indicates heartbeat success
"DeviceID" : "1c11c9c4-8dbc-4391-0dd7-56764ba18dbc"	Consistent with the information reported by the equipment foundation		
}			

5.3 ANPR picture and data push

URI	/NotificationInfo/TollgateInfo			
Description	Push ANPR snapshot data and picture , which can support original drawing, license plate drawing and car drawing.			
Method	URI Request Parameters	Request message	Response message	Remarks

POST	None	Refer to ANPR data Push message body determination	Refer to ANPR reply response Definition	
Notes				

ANPR data push message body definition:

Sample	Description	Mandatory	Value
"Picture":{	Picture information		
"NormalPic" : {	Original drawing	No	
"PicName" : "xxx.jpg",	Picture name		
"Content" : ""	Picture data		Base64 encoded data
},			
"CombinPic": {[	Combin drawing	No	Array
"PicName" : "xxx.jpg",	Picture name		
"Content" : ""	Picture data		Base64 encoded data
],...			
},			
"CutoutPic ":{	License plate matting	No	
"PicName" : "xxx.jpg",	Picture name		
"Content" : ""	Picture data		Base64 encoded data
},			
"VehiclePic":{	Vehicle drawing	No	
"PicName" : "xxx.jpg",	Picture name		
"Content" : ""	Picture data		Base64 encoded data
},			
"FacePic ":{[	Face drawing	No	Array
"PicType" : 1,	Picture type		int(32) Face Picture Type, See appendix for specific value classification APPENDIX 1.7 FACE TYPE DEFINITION
"PicName" : "xxx.jpg",	Picture name		
"Content" : ""	Picture data		Base64 encoded data
],...			
},			
"Plate":{	Plate Info		
"IsExist": true,	Is there a license	Yes	Bool

	plate		
"PlateNumber": "浙 A1111",	Plate number	No	char(64) Support multiple encoding methods: UTF-8, GB2312, hexadecimal ASCII encoding
"ADR": "浙 A1111",	Dangerous chemical vehicle number plate	No	char(32), Only when it is a dangerous chemical vehicle, it can be carried with ADR identification.
"PlateColor": "Blue",	Plate Color	Yes	See appendix for specific value classification APPENDIX 1.1 PLATECOLOR VALUE CLASSIFICATION
"PlateType": "Normal",	License plate type	Yes	See appendix for specific value classification APPENDIX 1.2 PLATETYPE VALUE CLASSIFICATION
"Confidence": 80,	Confidence	No	Confidence range 0-255
"BoundingBox": [10, 20, 30, 40],	License plate coordinates	Yes	Left, up, right, down
"Country": "",	Country of license plate	No	char(32) v1.04 version delete this field
"UploadNum": "",	Upload serial number	No	int (32) Picture self-added serial number, 0, 1, 2 ..., etc.
"Channel": 0	Channel	No	int(32) Camera: There is only one channel, and the channel number is 0; Smart box: the channel number starts from 0; That is, channel number 0 corresponds to channel 1, channel number 1 corresponds to channel 2, etc.
"Region": "",	License plate ownership	No	char(32) It shall comply with iso3166 standard
},			

"Vehicle":{	Vehicle information		
"VehicleColor": "White"	Vehicle Color	No	See appendix for specific value classification Appendix 1.3 VehicleColor Values Classification
"VehicleSign": "Audi"	Car logo	No	See appendix for specific value classification Appendix 1.4 VehicleSign Value Classification
"VehicleType": "PassengerCar",	Vehicle type	No	See appendix for specific value classification Appendix 1.5 VehicleType Value Classification
"VehicleSeries": "UnKnown",	Vehicle Series		Temporarily unidentified, equipment temporarily unidentified Car system, followed by identification car system, and the equipment directly transmits the identification result
"Speed": 0	Speed	No	unit: km/h
"VehicleBoundingBox": [10, 20, 30, 40],	Body coordinates	No	Left, up, right, down
},			
"SnapInfo":{	Capture information		
"TriggerSource": "Coil",	Trigger source	No	"Unknown" "Coil" "Radar" "Video"
"SnapTime": "2018-03-01 16:00:00",	Capture time	No	yyyy-mm-dd hh:mm:ss, localTime of device
"AccurateTime": "2020-08-29 13:38:23.246",	Accurate time of capture	No	yyyy-mm-dd hh:mm:ss.sss, localTime of device Note: Accurate capture time, accuracy to milliseconds, new field in V1.03 version
"TimeZone": 0	TimeZone	No	Reference to Appendix1.8
"DSTTune": 0	DSTTune	No	DST Time Tune 1:the current time belongs

			to the DST 0: the current time is not the DST Time
"SnapAddress":",	Capture address	No	char(64)
"LanNo":1,	Lane number	No	New field in V1.07 version; int(32) lane number start with 1
"Direction": "Obverse",	Bayonet direction	No	"Obverse": Forward "Reverse": Reverse
"OpenStrobe":true,	Whether to open the gate		Bool type, ture:open strobe, false:don't open strobe
"AllowUser":true,	Is it a allowed list	No	Bool Type, Ture: Yes, False: No Note:Since V1.02, the field name has been adjusted to "AllowUser", and the old field is no longer compatible
"AllowUserEndTime": "2019-11-29 23:59:59"	Allow list expiration time	No	yyyy-mm-dd hh:mm:ss Note: Since V1.02, the field name has been adjusted to "AllowUserEndTime", and the old field is no longer compatible
"BlockUser":true,	Is it a forbidden list	No	Bool type,ture:yes, false:no Note:SinceV1.02,the field name has been adjusted to "BlockUser", and the old field is no longer compatible
"BlockUserEndTime": "2019-11-29 23:59:59"	Expiration time of prohibited list	No	yyyy-mm-dd hh:mm:ss Note: Since V1.02, the field name has been adjusted to "BlockUserEndTime", and the old field is no longer compatible
"DefenceCode": "xxxx"	Anti-counterfeiting code	Yes	
"DeviceID": 1c11c9c4-8dbc-4391-0dd7-56764ba18dbc"	Unique identification of the device,	Yes	Default using UUID

	which is the same as Consistency of reported information of equipment foundation		
"InCarPeopleNum":2	Number of people in the car	No	int(32) Number of people in the car captured and identified
}			
}			

Definition of checkpoint ANPR reply response

Sample	Description	Mandatory	Value
{	ANPR Response		
"Result" : true	Result status	Yes	true: Processed successfully false: Processing failed
"DeviceID": 1c11c9c4-8dbc-4391-0dd7-56764ba18dbc"	Unique identification of the device, which is the same as Consistency in basic information reporting	Yes	UUID
"PlateNumber" : "A12345",	Represents a successfully received License plate number	No	Char(32) Actual license plate number, string type
"AllowUser" : true	Is the license plate on the allowed list	No	true:allowlist false:not in allowlist If the application scenario does not require this function, you can either take this field without it or take the default value of true Note: Since V1.02, the field name has been adjusted to "AllowUser".Old fields are no longer compatible
"BlockUser" : true	Is the license plate on a prohibited list	No	true:Blocklist false:Not in Blocklist

			If the application scenario does not require this function, you can either take this field without it or take the default value of true Note: Since V1.02, the field name has been adjusted to "BlockUser". Old fields are no longer compatible
"OpenStrobe" : true	Whether to open the gate	No	true: Indicates that the gate needs to be opened false: Indicates that the gate will not be opened If the application scenario does not require this work, Yes, you may not carry this field
}			

5.4 Parking information and picture push

URI	/NotificationInfo/ParkingInfo			
Description	Push parking capture information and picture data, which can support original drawing, license plate matting and car body Graphs and composite graphs			
Method	URI Request Parameters	Request message	Response message	Remarks
POST	None	Refer to Parking Data Push Message Size Righteousness	Refer to Parking response definition	
Notes				

Parking Data Push Message Body Definition

Sample	Description	Mandatory	Value
"Picture":{	Picture information		
"NormalPic" : {	Original drawing	No	
"PicName" : "xxx.jpg",	Picture name		
"Width":1920	Picture resolution		
"Height":1080	Picture resolution		
"Content" : ""	Picture data		Base64 encoded data

"PicName_2nd" : "xxx_2.jpg",	Second Picture name		
"Width_2nd":1920	Second Picture resolution		
"Height_2nd":1080	Second Picture resolution		
"Content_2nd" : ""	Second Picture data		Base64 encoded data
},			
"CombinPic": { [	Combin drawing	No	Array
"PicName" : "xxx.jpg",	Picture name		
"Content" : ""	Picture data		Base64 encoded data
],...			
},			
"VehiclePic":{	Vehicle picture	No	
"PicName" : "xxx.jpg",	Picture name		
"Content" : ""	Picture data		Base64 encoded data
},			
"Plate":{	Plate information		
"IsExist": true,	Has a plate	Yes	Bool
"PlateNumber": "A1111",	Plate number	No	char(64) Support multiple encoding methods: UTF-8, GB2312, hexadecimal ASCII encoding
"PlateColor": "Blue",	Plate color	Yes	See appendix for specific value classification Appendix 1.1 PlateColor Value Classification
"PlateType": "Normal",	Plate type	Yes	See appendix for specific value classification APPENDIX 1.2 PLATE TYPE VALUE CLASSIFICATION
"Confidence": 80,	Confidence	No	Confidence range 0-100
"BoundingBox": [10, 20, 30, 40],	Plate coordinates	Yes	Upper left and lower right coordinate points
"Region": "",	License plate ownership	No	char(32) It shall comply with iso3166 standard
},			
"Vehicle":{	Vehicle information		
"VehicleColor": "White"	Vehicle Color	No	See appendix for specific value classification

			Appendix 1.3 VehicleColor Values Classification
"VehicleSign": "Audi"	Logo(Brand)	No	See appendix for specific value classification Appendix 1.4 VehicleSign Value Classification
"VehicleType": "PassengerCar",	Vehicle type	No	See appendix for specific value classification Appendix 1.5 VehicleType Value Classification
"VehicleSeries": "UnKnown",	Vehicle Series	No	Temporarily unidentified, equipment temporarily unidentified Car system, followed by identification car system, set Prepare for direct transmission of recognition results
"VehicleBoundingBox": [10, 20, 30, 40],	Body coordinates	No	Left, up, right, down
},			
"ParkingInfo": {	Capture information		
"SnapTime": "2020-03-01 16:00:00",	Shot time	Yes	yyyy-mm-dd hh:mm:ss
"TimeZone": 0	TimeZone	No	Reference to Appendix 1.8
"DSTTune": 0	DSTTune	No	DST Time Tune 1: the current time belongs to the DST 0: the current time is not the DST Time
"Channel": 0	Channel Number	No	int(32) Camera: There is only one channel, and the channel number is 0; Smart box: the channel number starts from 0; That is, channel number 0 corresponds to channel 1 and channel Number 1 corresponds to channel 2..., etc.
"ParkingStallsNo": "",	Parking space number	Yes	char(64), uniformly using UTF-8 encoding. If not recognized, the field is

			empty String
"Direction": "Obverse",	Parking direction	No	"Obverse": Forward "Reverse": Reverse "Unknow": Direction not recognized
"ParkingStatus": 0,	Parking state	YES	Int(32), 0: The vehicle stops in the parking space, 1: The vehicle leaves the parking space, 2: Unknown state 3: Illegal parking 4: Parking detection
"AllowUser": true,	Is it a allowed list	No	Bool
"BlockUser": true,	Is it a forbidden list	No	Bool
"Relationship": [[	Capture information of entering vehicles associated with vehicle appearance capture	C	It is only carried when the captured information is reported when the vehicle comes out. Is an array, there may be matches to multiple vehicles entering Record;
"Confidence": 80,	confidence	No	The value range is 0-100
"ParkingStallsNo": "",	Parking space number	Yes	char(64), unified use of UTF-8. When not recognized, the field is an empty string
"PlateNumber": "A1111",	Plate number	No	char(32)
],...			
}			
"DeviceID": 1c11c9c4-8dbc-4391-0dd7-56764ba18dbc"	Unique identification of the device, which is the same as Consistency of reported information of equipment foundation	C	Default using UUID , which needs to be carried when supporting the basic information push function;
}			

"inRecordId": "Nc664JUwkfCQib4E"	Entry record id	YES	String (only exists when leaving the site to match the entry record)
}			

Parking response definition

Sample	Description	Mandatory	Value
{	Parking response		
"Result" : true		Yes	true: Successfully received push; false: Message processing failed for reasons described in Message
"Message" : "Success"		Yes	Message processing result description
"RspTime" : "2020-08-11 16:00:01"		Yes	Reply time after message processing
}			

5.5 Abnormal alarm information push

URI	/NotificationInfo/AlarmInfo			
Description	Push the relevant abnormal situations detected in the parking business scenario, which is convenient for the platform to identify and pass Know manual intervention confirmation			
Method	URI Request Parameters	Request message	Response message	Remarks
POST	None	Refer to Alarm Data Push Message definition	Refer to Alarm reply response definition	
Notes				

Alarm data push message definition:

Sample	Description	Mandatory	Value
"AlarmInfo":{	Alarm information		
"Time": "2020-08-11 16:00:00",	Alarm time	Yes	yyyy-mm-dd hh:mm:ss
"Type" : 0	Alarm type	Yes	int(32) See appendix for specific value classification

			Appendix 1.6 Alarm Type Definition
"State": "Pluse"	Alarm state	No	This field is new added in V1.07 version: "Pluse": Pluse event "Start": event start "Stop": event stop
"Confidence": 80,	Matching confidence	No	The value range is 0-100
"ParkingStallsNo": "",	Parking space number	No	char(64), uniformly using UTF-8 encoding. If not recognized, the field is empty String
"PlateInfoList": [[	plate number information	No	Is an array, when the alarm type is 0, There may be multiple license plates.
"PlateNumber": "A1111",	plate number	No	char(64) Support multiple encoding methods: UTF-8, GB2312, hexadecimal ASCII encoding
"EntryTime": "2020-08-11 16:00:00"	Admission time of the license plate car	C	yyyy-mm-dd hh:mm:ss When the alarm type is "overtime parking" and other events, it is necessary to carry the entry time of the corresponding license plate.
"TimeZone": 0	TimeZone	No	Reference to Appendix 1.8
"DSTTune": 0	DSTTune	No	DST Time Tune 1: the current time belongs to the DST 0: the current time is not the DST Time
],...			
}			
"DeviceID": "1c11c9c4-8dbc-4391-0dd7-56764ba18dbc"	Consistent with the information reported by the equipment foundation	C	Default using UUID, which should be carried when basic information push function is supported
}			

Alarm Reply Response Definition:

Sample	Description	Mandatory	Value
{	Alarm Response		
"Result" : true		Yes	true: Successfully received push; false: Message processing failed for reasons described in Message
"Message" : "Success"		Yes	Message processing result description
"RspTime" : "2020-08-11 16:00:01"		Yes	Reply time after message processing
}			

5.6 E-police alarm information and data push

URI	/NotificationInfo/TrafficViolationInfo			
Description	Push the violation information and picture data of electric police, and can support the violation composite(FTP path or Base64 Picture)			
Method	URI Request Parameters	Request message	Response message	Remarks
POST	None	Refer to E-Police Violation Data Push Message Body definition	Violation reply response definition	
Notes				

Electric Police Violation Data Message Definition:

Sample	Description	Mandatory	Value
"Picture":{	Picture information		
"CombinPic" : {	Composite graph	No	
"PicPath" : "ftp://xxxx/abc.jpg",	Picture name	No	Path or Base64 data Choose one of the two formulas
"Content" : ""	Picture data	No	Base64 encoded data
},			
"Plate":{	License plate information		Only valid when the motor vehicle violates the rules at that time
"IsExist": true,	Has a license plate	Yes	Bool

"PlateNumber": "A1111",	plate number	No	char(64) Support multiple encoding methods: UTF-8, GB2312, hexadecimal ASCII encoding
"ADR": "A1111",	ADR plate number	No	char(32), only when it is a dangerous vehicle
"PlateColor": "Blue",	plate color	No	See appendix for specific value classification Appendix 1.1 PlateColor Value Classification
"PlateType": "Normal",	plate type	No	See appendix for specific value classification APPENDIX 1.2 PLATE TYPE VALUE CLASSIFICATION
"Confidence": 80,	Confidence	No	Confidence range 0-255
"BoundingBox": [10, 20, 30, 40],	Plate coordinates	C	Upper left, lower right
"UploadNum": "",	Serial number	No	int(32) Picture self-added serial number, 0, 1, 2 ..., etc.
"Channel": 0	Channel number	No	int(32) Camera: There is only one channel, and the channel number is 0; Smart box: the channel number starts from 0; That is, channel number 0 corresponds to channel 1, channel number 1 corresponds to channel 2, etc.
"Region": "",	License plate ownership	No	char(32) It shall comply with iso3166 standard
},			
"Vehicle": {	Vehicle information		Only valid when the vehicle violates the regulations at that time

"LanNo":1	Lane number		int(32) Lane number of the vehicle, starting from 1
"VehicleColor": "White"	Vehicle Color	No	See appendix for specific value classification Appendix 1.3 VehicleColor Values Classification
"VehicleSign": "Audi"	Logo(Brand)	No	See appendix for specific value classification Appendix 1.4 VehicleSign Value Classification
"VehicleType": "PassengerCar",	Vehicle type	No	See appendix for specific value classification Appendix 1.5 VehicleType Value Classification
"VehicleSeries": "UnKnown",	Vehicle Series	No	Tentatively not identified, the equipment temporarily doesn't identify the vehicle system, and the subsequent identification of the vehicle system, setting Prepare for direct transmission of recognition results
"VehicleBoundingBox": [10, 20, 30, 40],	Body coordinates	No	Left, up, right, down
},			
"TrafficViolationInfo": {	Violation information		
"ViolationType": "TrafficRunRedLight",	Violation type	Yes	char(256) See appendix for specific value classification APPENDIX 1.9 VIOLATIONTYPE VALUE CLASSIFICATION
"ViolationCode": "1001",	Violation code	No	char(4)
"AccurateTime": "2020-08-29 13:38:23.246",	Accurate time of capture	No	yyyy-mm-dd hh:mm:ss.sss Note: Accurate capture time, accuracy to milliseconds

"TimeZone":0	TimeZone	No	Reference to Appendix1.8
"DSTTune":0	DSTTune	No	DST Time Tune 1:the current time belongs to the DST 0: the current time is not the DST Time
"DefenceCode": "xxxx"	Anti-counterfeiting code	Yes	
"DeviceID": 1c11c9c4-8dbc-4391-0dd7-56764ba18dbc"	Unique identification of equipment, and reporting information of equipment foundation Consistency of	C	Default using UUID, which needs to be carried when using the basic information push in 5.1
}			
}			

Electric police reply message:

Sample	Description	Mandatory	Value
{	Electric police reply message		
"Result" : true	processing results	Yes	bool: true: succeeded false: failed
"DeviceID": 1c11c9c4-8dbc-4391-0dd7-56764ba18dbc"	Unique identification of equipment, and the basic information report Consistent	C	When using the basic information push in 5.1, this field needs to carry
"Message" : "OK",	Description of processing results	No	Description of processing results
}			

5.7 Traffic flow statistical information push

URI	/NotificationInfo/TrafficFluxInfo
Description	Push traffic flow statistics, which can support lane, time period traffic flow, vehicle queuing, Lane occupancy, headway, lane status, etc.

Method	URI Request Parameters	Request message	Response message	Remarks
POST	None	Refer to traffic flow Push message definition	Refer to traffic flow response definition	
Notes				

traffic flow Push message definition:

Sample	Description	Mandatory	Value
{	traffic flow		
"DeviceID": 1c11c9c4-8dbc-4391-0dd7-56764ba18dbc"	Unique identification of the device, which is the same as Consistency in basic information reporting	C	Default using UUID, Whenusing the basic information push in 5.1, this field needs to carry
"CaptureInterval":30	Acquisition cycle	Yes	int Traffic statistics period, unit second (s)
"TrafficFluxInfo" :[{	Flow statistics	No	The array subscript corresponds to the lane number, Index 0 corresponds to lane 1, and so on
"AverLaneFluxCount":28.123	Average lane flow data	No	Double The traffic volume ofthe lane in the statistical period
"LaneVelocity":30.56	Average lane speed	No	Double Unit: km/h
"LaneHandRate":0.45	Average lane occupancy rate	No	Double The value range is between 0 and 1
"AverHeadSpace":2.30	Average lane headway	No	Double: Unit: Meter
"AverHeadTime":3.00	Average lane headtime	No	Double: Unit: Seconds
"AverQueueLen":10.45	Average queuing length of lane	No	Double: Unit: Meter
"LanBlockState":0	Lane congestion state	No	Int(32): 0: Smooth 1: Congestion
],...			

}			
}			

Flow statistics reply message definition:

Sample	Description	Mandatory	Value
{	Flow statistics reply response		
"Result" : true	processing results	Yes	true: Processed successfully false: Processing failed
"DeviceID": "1c11c9c4-8dbc-4391-0dd7-56764ba18dbc"	Unique identification of the device, which is the same as Consistency in basic information reporting	C	When using the basic information push in 5.1, this field needs to carry
"Message" : " Received successfully ",	Description of processing results	No	Description of processing results
}			

5.8 Timed parking space status push

URI	/NotificationInfo/TimedParkingSpaceInfo			
Description	Periodically push parking space status information, support parking space number, parking space type, etc			
Method	URI Request Parameters	Request message	Response message	Remarks
POST	None	See definition of timed parking space push message body	Definition of timed parking space status reply response	
Notes				

Definition of timed parking space status data push message:

Sample	Description	Mandatory	Value
{	Timed parking space push information		
"DeviceID": "1c11c9c4-8dbc-4391-0dd7-56764ba18dbc"	Unique identification of the device, which is	C	Default using UUID, When using the basic information push in 5.1, this

	the same as Consistency in basic information reporting		field needs to carry
"TimingPeriod":10	Timing cycle	Yes	Int(32) Regular push cycle of parking space status, unit second (s)
"Time": "2020-08-11 16:00:00",	Time of occurrence	Yes	yyyy-mm-dd hh:mm:ss
"TimeZone":0	TimeZone	No	Reference to Appendix1.8
"DSTTune":0	DSTTune	No	DST Time Tune 1:the current time belongs to the DST 0: the current time is not the DST Time
"ParkingSpacePic":{	Panorama of parking space	No	
"PicName" : "xxx.jpg",	Picture name		
"Content" : ""	Picture data		Base64 encoded data
},			
"TimedParkingSpaceInfo":[{	Parking space statistics	Yes	Array Each element represents a parking space
"ParkNo": "C001"	Parking space No		char[32]
"SpaceType":0	Parking space type		Int(32) 0 : Trolley parking space, 1 : Cart parking space 2 : Parking space for hazardous chemicals, 3 : Truck parking space, 4: Passenger car parking space, 5: Bus parking space
"Used":true	Is there a car in the parking space		Bool
],...			
}			
}			

Definition of timed parking space status reply message:

Sample	Description	Mandatory	Value
{	Timed parking space push		

	response		
"Result" : true	processing results	Yes	true: Processed successfully false: Processing failed
"DeviceID": 1c11c9c4-8dbc-4391-0dd7-56764 ba18dbc"	Unique identification of the device, which is the same as Consistency in basic information reporting	C	When using the basic information push in 5.1, this field needs to carry
"Message" : " Received successfully ",	Description of processing results	No	Description of processing results
}			

Appendix A

Appendix 1.1 PlateColor Value Classification

"White|Black|Red|Yellow|Gray|Blue|Green|Pink|Purple|Darkviolet|Brown|Maroon|Silver|Dimgray|Whitesmoke|Darkorange|Mistyrose|Tomato|Olive|Gold|Darkolivegreen|Chartreuse|Greenyellow|Forestgreen|Seagreen|Deepskyblue|Cyan|DarkBlue|DarkRed|DarkGreen|DarkYellow|DarkPink|DarkPurple|DarkBrown|DarkCyan|Orange|DarkGold|ShadowGreen|YellowGreen"

Appendix 1.2 PlateType Value Classification

"Yellow|Normal|Embassy Car|Embassy|OverSea Car|Foreign Car|Police|Armed|Military|DoubleMilitary|DoubleYellow|Training|Personal|Agri|Moto|Tractor|Black|Red|Cyan|White|PureNewEnergyMicroCar|MixedNewEnergyMicroCar|PureNewEnergyLargeCar|MixedNewEnergyLargeCar|Other"

Appendix 1.3 VehicleColor Values Classification

"White|Black|Red|Yellow|Gray|Blue|Green|Pink|Purple|Darkviolet|Brown|Maroon|Silver|Dimgray|Whitesmoke|Darkorange|Mistyrose|Tomato|Olive|Gold|Darkolivegreen|Chartreuse|Greenyellow|Forestgreen|Seagreen|Deepskyblue|Cyan|DarkBlue|DarkRed|DarkGreen|DarkYellow|DarkPink|DarkPurple|DarkBrown|DarkCyan|Orange|DarkGold|ShadowGreen|YellowGreen"

Appendix 1.4 VehicleSign Value Classification

"Other|Audi|Honda|Buick|Volkswagen|Toyota|BMW|Peugeot|Ford|Mazda|Nissan|Hyundai|Suzuki|Citroen|Benz|BYD|Geely|Lexus|Chevrolet|Chery|Kia|Charade|DF|Naveco|SGMW|Jinbei|JAC|Geely|ChangAn|Great Wall|Skoda|BaoJun|Subaru|LandWind|Luxgen|Renault|Mitsubishi|Roewe|Cadillac|MG|Zotye|ZhongHua|Foton|SongHuaJiang|Opel|HongQi|Fiat|Jaguar|Volvo|Acura|Porsche|Jeep|Bentley|Bugatti|ChuanQi|Daewoo|DongNan|Ferrari|Fudi|Huapu|HawTai|JMC|JingLong|JoyLong|Karry|Chrysler|Lamborghini|RollsRoyce|Linian|LiFan|LieBao|Lincoln|LandRover|Lotus|Maserati|Maybach|Mclaren|Youngman|Tesla|Rely|Lsuzu|Yiqi|Infiniti|YuTong|AnKai|Canghe|HaiMa|Crown|HuangHai|JinLv|JinNing|DaoQi|Europestar|MINI|Gleagle|ShiDai|ShuangHuan|TianYe|WeiZi|Englon|ZhongTong|Changan|Yuejin|Taurus|Alto|Weiwang|Chenglong|Haige|Shaolin|Beifang|Beijing|Hafu|Yongyuan|Shangqidatong|beiqizhizao|YEMA|GONOW|Hummer|Zhongshun|SSANGYONG|hangtianchenggong|Koenigsegg|Aston

Martin|TechArt|Carlsson|SEAT|Wiesmann|AlfaRomeo|SPYKER|Mercury|Scania|PROTO
N|LANCIA|Panoz|Holden|ASCARI|Dacia|Renault|Samsung|Vauxhall|Venturi|Morgan|Hin
o|PONTIAC|KarlAbarth|Saturn|Tianma|DAIHATSU|Oldsmobile|Saibao|Moskvich|DS|Ven
ucia|MITSUOKA|GMG|Polonez|Jiangnan|LADA|Scion|DAF|Yizhing|Xinkai|Dadi|TataMot
ors|KAMAZ|Datsun|Foden|HerbertAustin|ronart|Hengtian|DENZA|ZINORO|Fornasari|Fu
qiQiteng|QOROS|HUASONG|COWIN|LEVDEO|Lingzhi|GreenWheelEV|Nanjingjinlong|
RUF|Shanqitongjia|Siming|Yujie|ZD|BAC|Borgward|Conquest|DMC|Elemental|GAZ|GM
C|KTM|nanoflowcell|noble|ssc|Tramontana|zenvo|BRABUS|Patton|MELKUS|BQHS|AY|
RIICHI|FuShiDa|ChuanQiYeMa|HeiBao|NanJun|HuaLing|SINOTRUK|CCTRUCKS|SANY
|KAMA|BEIBENTRUCK|YTC|HongYan|TKNG|YangZi|SINOMOTORS|YangZhouYaXing|
YouYiKeChe|ELFIN|JOSS|MAN|ALPHINA|RAV4|Ascari|Allard|Bitter|Bizzarrini|Pininfarina
|PGO|Bertone|Bistol|DeTomaso|delage|delahaye|diatto|nsu|SCANIA|Farboud|Mustang|
GAZZ|Horch|Caterham|Corvette|Lagonda|Leyland|RUFU|MGROVER|RochetSchneider|
Marcos|MACK|MEILU|metropolitan|MeiYa|Plymouth|italdisign|TATRA|Trabant|TVR|Wart
burg|cizeta|Shelby|Cisitalia|Seata|DKW|EDAG|Ikarus|UAZ|ZOEMO|JMC|Kangdi|Saab|S
hifeng|Aochi|Yaolong|Dongfanghong|Sanhuanshitong|Xugong|Chongqiawangpai|Zhongg
uozhongqi|Feichi|Guangtong|Mudang|BAC|BAC|Shengwo|Xiwo|Wuzhoulong|Tongxin|Ha
fu|Jinggong|Jianghuan|Chunlan|Dayun|LiFan|Shanqi|Naveco|LiYaNa|ChangAn|ChangA
n|Hyundai|Hino|Hino|JAC|JAC|JMC|Baowo"

Appendix 1.5 VehicleType Value Classification

"Unknown|PassengerCar|LargeTruck|MidTruck|SaloonCar|Microbus|MicroTruck|Tricycle|
Motor|Passerby|SUV-MPV|MidPassengerCar|TankCar|SUV|MPV|Bus|Pickup|MiniCarria
ge|OilTankTruck|DregsCar|ConcreteMixerTruck"

Appendix 1.6 Alarm Type Definition

Alarm Type	Description
0	ParkingOverload
1	AcrossParking
2	Reverse parking
3	Timeout parking
4	Lack of parking
5	Parking without license
6	NonMotor Parking
7	Illegal parking
8	No Parking
9	Blocklist parking
10	Parking spot inspection

11	Trolley occupying large parking space
12	Non new energy vehicles occupy new energy parking spaces
13	Retention alarm
14	Barrier open
15	The barrier is not fully opened
16	Barrier closed
17	The barrier is not fully closed
18	Video Occlusion
19	Video Occlusion Recovery
99	Other

Appendix 1.7 Face Type Definition

FaceType	Description
0	Other
1	Driver's face
2	Co pilot's face

Appendix 1.8 TimeZone Definition

TimeZone	TimeZone Name	Deviation From UTC-0 (Second)
0	GMT+00:00	0*3600
1	GMT+01:00	1*3600
2	GMT+02:00	2*3600
3	GMT+03:00	3*3600
4	GMT+03:30	3*3600+1800
5	GMT+04:00	4*3600
6	GMT+04:30	4*3600+1800
7	GMT+05:00	5*3600
8	GMT+05:30	5*3600+1800
9	GMT+05:45	5*3600+1800+900
10	GMT+06:00	6*3600
11	GMT+06:30	6*3600+1800

12	GMT+07:00	7*3600
13	GMT+08:00	8*3600
14	GMT+09:00	9*3600
15	GMT+09:30	9*3600+1800
16	GMT+10:00	10*3600
17	GMT+11:00	11*3600
18	GMT+12:00	12*3600
19	GMT+13:00	13*3600
20	GMT-01:00	-1*3600
21	GMT-02:00	-2*3600
22	GMT-03:00	-3*3600
23	GMT-03:30	-3*3600-1800
24	GMT-04:00	-4*3600
25	GMT-05:00	-5*3600
26	GMT-06:00	-6*3600
27	GMT-07:00	-7*3600
28	GMT-08:00	-8*3600
29	GMT-09:00	-9*3600
30	GMT-10:00	-10*3600
31	GMT-11:00	-11*3600
32	GMT-12:00	-12*3600
33	GMT-4:30	-4*3600-1800
34	GMT+10:30	10*3600+1800
35	GMT+14:00	14*3600
36	GMT-09:30	-9*3600-1800
37	GMT+08:30	8*3600+1800
38	GMT+08:45	8*3600+2700
39	GMT+12:45	12*3600+2700

Appendix 1.9 ViolationType Value Classification

"QSYTrafficCarWeight|TrafficOverSpeed|Overspeed|TrafficUnderSpeed|UnderSpeed|TrafficRunRedLight|RunRedLight|TrafficRetrograde|Retrograde|TrafficOverLine|Overline|TrafficWrongRoute|TrafficJam|TrafficOverYellowLine|TrafficCrossLane|TrafficU-Turn|TrafficVehicleInRoute|TrafficYellowPlateInLane|TrafficTurnLeft|TrafficTurnRight|TrafficNoPassing|TrafficVehicleInBusRoute|TrafficBacking|TrafficOverStopLine|TrafficPedestrianPriority|TrafficParkingSpaceOverLine|TrafficRestrictedPlate|TrafficParkingOnYellowBox|TrafficStay|TrafficWithoutSafeBelt|TrafficDriverSmoking|TrafficDriverCalling|TrafficPedestrianRunRedLight|TrafficJamForbidInto|TrafficPassNotInOrder|TrafficThrow|TrafficPedestrian|SmokeDetection|TrafficNonMotorWithoutSafehat|TrafficNonMotorOverload|TrafficNonMotorInMotorRoute|TrafficTruckforbid|TrafficHighBeam|TrafficTurnRightAfterPeople|TrafficTurn

RightAfterStraight|TrafficQueueJump|TrafficBigBendSmallTurn|TrafficTurnLeftAfterStraight"

Appendix B

Appendix Table 1-1 Revision record

No.	Brief introduction of revised contents	Reviser	Revision Date	Version
1	Version creation, including applicable scenarios, protocol framework, authentication and infrastructure For information push, picture and information push, etc.	Yangbin3	2020/3/27	V1.00
2	Add ANPR picture information: upload serial number and channel number	Panguangying	2020/5/15	V1.01
3	Add functional agreements related to berth business,including attributes and pictures ofvehicles entering and leaving berths, abnormal alarm functions, etc. Modification ofthe Allowed and Prohibited List Agreement The field names are Allow and Block; Adjusttherange of confidence parameters;	yangbin3	2020/8/10	V1.02
4	Increasing the millisecond accuracy of snapshot time in SnapInfo in ANPR data push FieldAccurateTime	yangbin3	2020/8/29	V1.03
5	ADR license plate field is added for dangerous vehicles, and the protocol description ofpermitted and prohibited lists is adjusted. The old fields are no longer compatible. Delete Country in bayonet message Field	yangbin3	2020/11/10	V1.04
6	Add the functions of pushing illegal information of electric police and pushing statistical information oftraffic flow	yangbin3	2020/11/16	V1.05
7	Alarm type adds parking space status inspection	panguangying	2020/12/10	V1.06
8	Increasing the Lane number in ANPR data push, and increasing the status of alarm in alarm infoemation push.	yangbin3	2021/7/19	V1.07
9	The new license plate number supports multiple coding methods: UTF-8, GB2312, hexadecimal ASCII coding	zhaoxue2	2021/08/25	V1.08
10	5.4 new fields are added to the parking interface, and 2 original drawings can be	panguangying	2021/09/08	V1.09

	pushed			
11	The bayonet information push is added to support the push of face pictures; New face image type definition table	panguangying	2021/11/03	V1.10
12	New value of alarm type mapping table	panguangying	2021/11/24	V1.11
13	5.4 parking interface parking status value added	panguangying	2022/01/06	V1.12
14	5.3、5.4、5.5 and 5.6 increase the current time zone index and daylight-saving time flag	maoxiangying	2022/02/10	V1.13
15	5.3 new number of people in the vehicle at bayonet interface	panguangying	2022/03/16	V1.14
16	Adjust the description of deviceid and add the violation type field in 5.6	panguangying	2022/04/19	V1.15
17	5.3, 5.4 and 5.6 add the license plate ownership field in the interface	panguangying	2022/05/06	V1.16
18	5.3 and 5.4 new composite diagram of interface	panguangying	2022/07/06	V1.17
19	5.3 and 5.4 and 5.6 add body coordinates of interface	zhaoxue	2022/08/16	V1.18
20	5.5 interface extension alarm status type definition	panguangying	2022/08/29	V1.19
21	Add 5.8 push interface of timed parking space status	panguangying	2022/09/13	V1.20